

Basic Information

Experiment No: 05

Date: 23/09/2026

Title: Student Performance Analysis using Classes, Arrays and Methods

Objective / Aim

To design and implement a Student Performance Analysis System in Java using classes, objects, arrays, constructors, methods, strings, loops, and conditional statements to calculate total marks, average, highest mark, lowest mark, percentage, grade, and result.

System Requirements

Hardware

- Standard PC/Laptop
- Intel Core i3/i5 processor or equivalent
- Minimum 4 GB RAM
- Minimum 500 MB free storage

Software

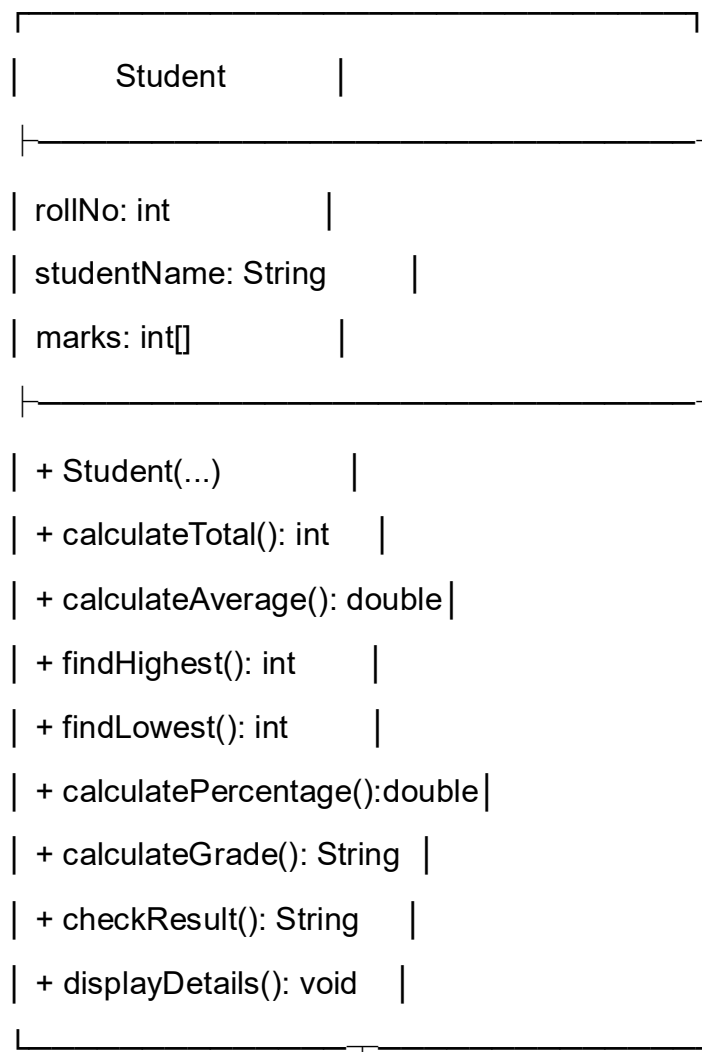
- Operating System: Windows / Linux / macOS
- Java Development Kit (JDK 17 or JDK 21)
- Text Editor or IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code
- Command Prompt / Terminal

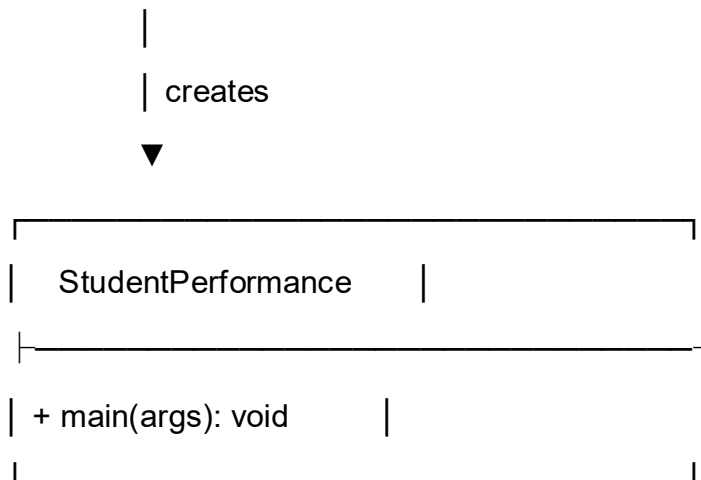
Algorithm / Logic Description

1. Create a "Student" class with roll number, name, and marks.

2. Accept marks for five subjects using an array.
3. Calculate total, average, highest, lowest, and percentage.
4. Assign a grade based on the percentage.
5. Check whether the student has passed or failed.
6. Display the student details and performance report.
7. Display a performance remark based on the grade.
8. End the program.

UML Class Diagram





Relationship: The "StudentPerformance" class creates a "Student" object and uses its methods to calculate and display the student's performance.

Source Code

```
package javacore;

import java.util.Scanner;

class Student {

    // Data members
    int rollNo;
    String studentName;
    int[] marks = new int[5];

    // Constructor
    Student(int rollNo, String studentName, int[] marks) {
        this.rollNo = rollNo;
        this.studentName = studentName;
        this.marks = marks;
    }
}
```

```
}
```

```
// Calculate total marks
```

```
int calculateTotal() {
```

```
    int total = 0;
```

```
    for (int i = 0; i < marks.length; i++) {
```

```
        total = total + marks[i];
```

```
    }
```

```
    return total;
```

```
}
```

```
// Calculate average marks
```

```
double calculateAverage() {
```

```
    return calculateTotal() / 5.0;
```

```
}
```

```
// Find highest marks
```

```
int findHighest() {
```

```
    int highest = marks[0];
```

```
    for (int i = 1; i < marks.length; i++) {
```

```
        highest = Math.max(highest, marks[i]);
```

```
    }
```

```
    return highest;
```

```
}
```

```
// Find lowest marks

int findLowest() {
    int lowest = marks[0];

    for (int i = 1; i < marks.length; i++) {
        lowest = Math.min(lowest, marks[i]);
    }

    return lowest;
}

// Calculate percentage
double calculatePercentage() {
    double percentage =
        (calculateTotal() / 500.0) * 100;

    // Round to two decimal places
    return Math.round(percentage * 100.0) / 100.0;
}

// Assign grade
String calculateGrade() {

    double percentage = calculatePercentage();

    if (percentage >= 90) {
        return "A+";
    }

    else if (percentage >= 80) {
```

```
        return "A";
    }
    else if (percentage >= 70) {
        return "B";
    }
    else if (percentage >= 60) {
        return "C";
    }
    else if (percentage >= 50) {
        return "D";
    }
    else {
        return "F";
    }
}
```

// Check pass or fail

```
String checkResult() {

    if (calculatePercentage() >= 50) {
        return "PASS";
    }
    else {
        return "FAIL";
    }
}
```

// Display student details

```
void displayDetails() {
```

```

// String operations

String name =
    studentName.trim().toUpperCase();

System.out.println(
    "\n===== STUDENT PERFORMANCE REPORT ====="
);

System.out.println("Student Name : " + name);
System.out.println("Name Length : " + name.length());
System.out.println("Roll Number : " + rollNo);

System.out.println("\nSubject Marks:");

for (int i = 0; i < marks.length; i++) {
    System.out.println(
        "Subject " + (i + 1) + " : " + marks[i]
    );
}

System.out.println(
    "\nTotal Marks : " + calculateTotal()
);

System.out.println(
    "Average : " + calculateAverage()
);

```

```
System.out.println(  
    "Highest Mark : " + findHighest()  
);
```

```
System.out.println(  
    "Lowest Mark : " + findLowest()  
);
```

```
System.out.println(  
    "Percentage : " + calculatePercentage() + "%"  
);
```

```
System.out.println(  
    "Grade      : " + calculateGrade()  
);
```

```
System.out.println(  
    "Result     : " + checkResult()  
);
```

```
// Performance remark
```

```
String grade = calculateGrade();
```

```
if (grade.equals("A+")) {  
    System.out.println(  
        "Remark      : Excellent Performance"  
    );  
}  
else if (grade.equals("A")) {
```

```
        System.out.println(
            "Remark      : Very Good Performance"
        );
    }
    else if (grade.equals("B")) {
        System.out.println(
            "Remark      : Good Performance"
        );
    }
    else if (grade.equals("C")) {
        System.out.println(
            "Remark      : Average Performance"
        );
    }
    else if (grade.equals("D")) {
        System.out.println(
            "Remark      : Needs Improvement"
        );
    }
    else {
        System.out.println(
            "Remark      : Failed - Needs More Practice"
        );
    }
}
```

```
public class StudentPerformance {
```

```
public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    // Input student details
    System.out.print("Enter Roll Number: ");
    int rollNo = sc.nextInt();

    sc.nextLine();

    System.out.print("Enter Student Name: ");
    String studentName = sc.nextLine();

    // Store marks of five subjects
    int[] marks = new int[5];

    System.out.println("\nEnter marks for 5 subjects:");

    for (int i = 0; i < 5; i++) {

        System.out.print(
            "Subject " + (i + 1) + ": "
        );

        marks[i] = sc.nextInt();
    }

    // Create Student object
```

```
Student student =  
    new Student(rollNo, studentName, marks);  
  
    // Display complete report  
    student.displayDetails();  
  
    sc.close();  
}  
}
```

Compilation & Execution Commands

Save the source code as:

StudentPerformance.java

If you are using the "package javacore;" statement, save the file inside the appropriate "javacore" package/folder.

Compile the program:

```
javac StudentPerformance.java
```

Run the program:

```
java StudentPerformance
```

Output Screen

Example Output

Enter Roll Number: 102

Enter Student Name: Sameera

Enter marks for 5 subjects:

Subject 1: 96

Subject 2: 97

Subject 3: 98

Subject 4: 99

Subject 5: 93

===== STUDENT PERFORMANCE REPORT =====

Student Name: Sameera

Name Length: 7

Roll Number: 102

Subject Marks:

Subject 1: 96

Subject 2: 97

Subject 3: 98

Subject 4: 99

Subject 5: 93

Total Marks: 483

Average: 96.6

Highest Mark: 99

Lowest Mark: 93

Percentage: 96.6%

Grade: A+

Result: PASS

Remark: Excellent Performance

Result

The Student Performance Analysis System was successfully implemented in Java using classes, objects, constructors, arrays, methods, loops, string operations, and conditional statements. The program successfully calculates and displays the student's total, average, highest mark, lowest mark, percentage, grade, result, and performance remark.